

A GAP-TWO INEQUALITY FOR LINEAR EXTENSIONS AND A POLYNOMIAL BOUND FOR THE BALANCE CONSTANT OF POSETS OF BOUNDED RANGE

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ABSTRACT. Let f be a uniformly random linear extension of a finite poset P , and let $x \parallel y$. We show that $\mathbb{P}[f(x) - f(y) \geq 2] \geq \mathbb{P}[f(y) < f(x)]/(\pi_N + 1)$ whenever some element can lie between y and x . Here π_N is the largest number of elements incomparable to such an element. The constant is sharp for every value of π_N . We use this to replace the only exponentially small quantity in the proof of Theorem 1.3 of Aires, Chan, Pak and Panova [KSBFT]. The result: every finite non-chain poset P of range $\pi(P) \leq D$ satisfies

$$\delta(P) \geq C_{\text{BFT}} + \min\left(\theta_0, \frac{C_{\text{BFT}}}{(5 + 3\sqrt{5})(D + 1) + 1}\right), \quad C_{\text{BFT}} = \frac{5 - \sqrt{5}}{10}, \quad \theta_0 = 0.2764 - C_{\text{BFT}} \approx 6.80 \cdot 10^{-6}.$$

This replaces the bound $C_{\text{BFT}} + (D + 1)^{-4(D+1)}/4096$ of [KSBFT]. In particular $\delta(P) > 0.2764$ for every $D \leq 3471$.

1. STATEMENTS

Throughout, $P = (X, <)$ is a finite poset with $n = |X|$, and f is a uniformly random linear extension, viewed as a bijection $X \rightarrow [n]$. We write $u \parallel v$ when u, v are incomparable, $\pi(u) = \#\{v : v \parallel u\}$, and $\pi(P) = \max_u \pi(u)$ (the *range*). We set $h(u) = \mathbb{E}f(u)$, $\delta(P; u, v) = \min(\mathbb{P}[f(u) < f(v)], \mathbb{P}[f(v) < f(u)])$, and $\delta(P) = \max_{u \neq v} \delta(P; u, v)$.

Definition 1.1. For $x \parallel y$ put $N(x, y) = \{w \in X \setminus \{x, y\} : w \not\leq y \text{ and } x \not\leq w\}$ and $\pi_N = \max_{w \in N(x, y)} \pi(w)$.

Theorem 1.2 (Lemma W). *Let $x \parallel y$ in a finite poset.*

- (a) *Some linear extension g has $g(y) < g(w) < g(x)$ if and only if $w \in N(x, y)$. Hence $\mathbb{P}[f(x) - f(y) \geq 2] > 0$ if and only if $N(x, y) \neq \emptyset$.*
- (b) *If $N(x, y) \neq \emptyset$, then $\#\{g : g(x) = g(y) + 1\} \leq \pi_N \cdot \#\{g : g(x) = g(y) + 2\}$, and consequently*

$$\mathbb{P}[f(x) - f(y) \geq 2] \geq \mathbb{P}[f(x) = f(y) + 2] \geq \frac{\mathbb{P}[f(y) < f(x)]}{\pi_N + 1} \geq \frac{\mathbb{P}[f(y) < f(x)]}{\pi(P) + 1}.$$

- (c) *The constant is sharp: for every $D \geq 2$ there are posets with $\pi_N = \pi(P) = D$ and equality in (b). For $D \geq 3$ this holds even when (x, y, z) is a Case D BFT triple (Example 2.1).*

Theorem 1.3 (Theorem 1.3''). *Let $D \geq 2$ and let P be a finite poset that is not a chain, with $\pi(P) \leq D$. Put*

$$\theta_0 := 0.2764 - C_{\text{BFT}} = 6.79775 \dots \cdot 10^{-6}, \quad \theta_W(D) := \frac{C_{\text{BFT}}}{(5 + 3\sqrt{5})(D + 1) + 1}.$$

If $0 \leq \theta \leq \theta_0$ and $\theta < \theta_W(D)$, then $\delta(P) > C_{\text{BFT}} + \theta$. Consequently

$$\delta(P) \geq C_{\text{BFT}} + \min(\theta_0, \theta_W(D)),$$

with strict inequality when $\theta_W(D) > \theta_0$, which happens exactly for $D \leq D_0 = 3471$. For those D the bound reads $\delta(P) > 0.2764$. For $D > 3471$ it reads $\delta(P) \geq C_{\text{BFT}} + \theta_W(D) \approx C_{\text{BFT}} + 0.02361/(D + 1)$.

For comparison, [KSBFT, Thm. 1.3] gives $\delta(P) > C_{\text{BFT}} + (D + 1)^{-4(D+1)}/4096$.

What is imported. Theorem 1.2 is self-contained. Theorem 1.3 uses exactly the following four results from the literature, and nothing else. We state them in the form used. A BFT triple is a triple (x, y, z) of distinct elements, not forming a chain, with $h(x) \leq h(y) \leq h(z) \leq h(x) + 2$. Since $u \prec v$ implies $h(u) < h(v)$, the only possible comparabilities are $x \prec y$, $x \prec z$ and $y \prec z$. Up to duality and reversal of the triple, one of the following holds [BFT95, §4]: Case A ($x \prec z$, $y \prec z$, $x \parallel y$); Case B ($y \prec z$, $x \parallel y$, $x \parallel z$); Case C (pairwise incomparable); Case D ($x \prec z$, $x \parallel y$, $y \parallel z$).

Imported fact I1 (Kahn–Linial [KL91, p. 365]; general form [AK25a, Cor. 5.1(b)]; [KSFT, Lemma 2.1]). *If $h(x) \leq h(y)$ then $\mathbb{P}[f(y) < f(x)] \leq 1 - e^{-1}$.*

Imported fact I2 ([BFT95, Thms. 4.1, 4.2]). *In Cases A and B, $\mathbb{P}[f(y) < f(x)] \geq 1/3$.*

Imported fact I3 ([KSFT, Lemma 2.3 and Appendix A], from [BFT95, §6]). *In Case C, $\max(\delta(P; x, y), \delta(P; y, z)) > 0.2764$.*

Imported fact I4 ([BFT95, (5.2), (5.5), (5.7), (5.8)], as [KSFT, (2.1)–(2.2)]). *In Case D, put $p(i, j) = \mathbb{P}[f(y) - f(x) = i, f(z) - f(y) = j]$, $p = \mathbb{P}[f(y) < f(x)]$, $p' = \mathbb{P}[f(z) < f(y)]$, $S = p + p'$, $a_1 = p(1, 1)$, $a_2 = p(1, 2) + p(2, 1)$, $a_3 = p(2, 2)$. Then*

$$2a_1 + a_2 \leq S, \quad 2a_2 + 2a_3 \leq S, \quad 4a_1a_3 \leq a_2^2, \quad 7 \leq 9S + 7a_1 + 3a_2 + 3a_3.$$

Everything else from [KSFT] that we use is reproved below: Cor. 2.2, Lemma 2.4, Lemma 3.2 (with a weaker hypothesis and a new proof of (iii)), Lemma 4.1, and the identities (4.2)–(4.7). Neither Lemma 3.1 (the $q_m = (D+1)^{-m(D+1)}$ bound) nor Lemma 4.2 (the $M^2/441$ conversion) of [KSFT] is used. Lemma 4.2 is replaced by a linear conversion (Proposition 5.3), and the lower bound (4.11) on M by Proposition 5.2. Section 7 records our check of the proof of Fact I3.

2. PROOF OF LEMMA W

Proof of Theorem 1.2(a). If $g(y) < g(w) < g(x)$ then $w \not\prec y$ and $x \not\prec w$, so $w \in N(x, y)$. Conversely, let $w \in N(x, y)$. Adding the relations $y < w < x$ to $\prec$ creates a cycle only if $w \preceq y$, $x \preceq w$ or $x \preceq y$, and none of these holds. So the transitive closure is a partial order, and any of its linear extensions will do. Finally, $f(x) - f(y) \geq 2$ means some element sits strictly between y and x , and that element lies in $N(x, y)$. $\square$

Proof of Theorem 1.2(b). Let $E_k = \{g : g(x) = g(y) + k\}$ and $N = N(x, y) \neq \emptyset$. We define $\Phi : E_1 \rightarrow E_2$. Fix $g \in E_1$, with y at position i and x at position $i + 1$. Every element of N lies before y or after x . Put $A = \{w \in N : g(w) > g(x)\}$.

Case $A \neq \emptyset$. Let w be the first element of A in g , say $g(w) = i + 1 + j$ with $j \geq 1$. Let $t_0 = x, t_1, \dots, t_{j-1}$ be the elements in positions $i + 1, \dots, i + j$. We claim that each t_k is incomparable to w . First, $w \not\prec t_k$ because t_k precedes w in g . Suppose $t_k \prec w$. Then $k \neq 0$, because $x \not\prec w$. Also $t_k \not\prec y$, because t_k comes after y in g ; and $x \not\prec t_k$, since otherwise $x \prec t_k \prec w$. Hence $t_k \in N$ and t_k lies after x and before w , which contradicts the choice of w . Moving w to position $i + 1$ changes the relative order only of the incomparable pairs $\{w, t_k\}$. The result is therefore a linear extension, with y, w, x in positions $i, i + 1, i + 2$. Let $\Phi(g)$ be this extension.

Case $A = \emptyset$. Then N lies entirely before y . Let w be the last element of N in g , and let $s_1, \dots, s_{j-1}, s_j = y$ be the elements after w up to and including y . Again each s_k is incomparable to w . First, $s_k \not\prec w$ by position. Next, $w \not\prec y$ because $w \in N$. Finally, suppose $w \prec s_k$ with $k < j$. Then $s_k \not\prec y$, since otherwise $w \prec y$; $x \not\prec s_k$, because s_k precedes x ; and $s_k \neq x, y$. So $s_k \in N$ lies after w , which contradicts maximality. Moving w to just after y gives a linear extension in which y, w, x are consecutive. Let $\Phi(g)$ be this extension. The case split matters: if we moved the last N -element before y even when $A \neq \emptyset$, then Φ would not be a function of g .

Multiplicity. Fix $g' \in E_2$ and let w be its middle element, so $w \in N$. Any Φ -preimage of g' is obtained from g' by moving w in one of two ways:

- right by j places, past x and $j - 1$ further elements (the case $A \neq \emptyset$);
- left by j places, past y and $j - 1$ further elements (the case $A = \emptyset$).

So a preimage is determined by the side and by j . We showed above that in a preimage all j elements that w passes are incomparable to w . Let a_R and a_L be the lengths of the maximal runs of elements incomparable to w immediately to the right and to the left of w in g' . Then $j \leq a_R$ on the right side and $j \leq a_L$ on the

left side. If $y \prec w$, then $a_L = 0$, which only helps. The two runs are disjoint sets of elements incomparable to w , so

$$|\Phi^{-1}(g')| \leq a_R + a_L \leq \pi(w) \leq \pi_N.$$

Hence $|E_1| \leq \pi_N |E_2|$, and

$$\mathbb{P}[f(y) < f(x)] = \mathbb{P}[E_1] + \mathbb{P}[f(x) - f(y) \geq 2] \leq \pi_N \mathbb{P}[E_2] + \mathbb{P}[f(x) - f(y) \geq 2] \leq (\pi_N + 1) \mathbb{P}[f(x) - f(y) \geq 2].$$

Note that $\pi_N \geq 1$: an element $w \in N$ with $y \prec w \prec x$ would force $y \prec x$. $\square$

Example 2.1 (sharpness, Theorem 1.2(c)). *Generic family*, $D \geq 2$. Take a chain $x = c_1 \prec \dots \prec c_{D-1}$ and two further elements y, w incomparable to everything. Then $\pi(y) = \pi(w) = D = \pi(P)$ and $N(x, y) = \{w\}$. The event $f(y) < f(x)$ says that y is first among $\{y\} \cup \{c_i\}$, and then w can occupy any of $D + 1$ slots. Exactly one of these slots lies between y and x . So the ratio in (b) is $1/(D + 1)$.

Case D BFT triple, $D \geq 3$ (from the audit mg-e60e). Let Q_k be the $2+1$ poset $\{x \prec z\} \sqcup \{y\}$ with a chain $c_1 \prec \dots \prec c_k$ placed above all of it. Let $P_k = Q_k \sqcup \{w\}$ with w incomparable to everything, and put $D = k + 3$. Then $\pi(P_k) = \pi(w) = D$. In Q_k the heights are $(h(x), h(y), h(z)) = (4/3, 2, 8/3)$. Inserting w uniformly into the $D + 1$ slots multiplies every height by $(D + 2)/(D + 1) \leq 5/4$. So (x, y, z) is a Case D BFT triple: $h(z) - h(x) \leq 5/3$. Moreover $N(x, y) = \{w\}$, since z and every c_i lie above x . In Q_k , y before x forces y immediately before x . Hence $\mathbb{P}[f(y) < f(x)] = 1/3$ and $\mathbb{P}[f(x) - f(y) \geq 2] = 1/(3(D + 1))$, and the ratio is $1/(D + 1)$.

Remark 2.2 (relation to known inequalities). Let $a_i = \mathbb{P}[f(y) - f(x) = i]$ and $b_i = \mathbb{P}[f(x) - f(y) = i]$. Kahn and Saks [KS84] (as restated in [BFT95, Lemma 2.1]) prove $a_1 = b_1$ and the two-sided inequality $a_2 + b_2 \geq a_1 + b_1$. Lemma W is one-sided: $b_1 \leq \pi_N b_2$. It carries a range factor, which Example 2.1 shows cannot be replaced by an absolute constant. In the notation of [BFT95] it says $\epsilon = b_1/B \leq \pi_N/(\pi_N + 1)$, where $B = \sum_i b_i$. We have not found Lemma W in the literature (see the accompanying search log).

Remark 2.3. Lemma W bounds a *ratio*. Absolute probabilities of such events can be exponentially small in the range. For example, for two parallel k -chains the probability that the top of one precedes the bottom of the other is $\binom{2k}{k}^{-1}$. This is why [KSFT, Lemma 3.1], which bounds absolute probabilities, loses $(D+1)^{-m(D+1)}$. In the application below the denominator $\mathbb{P}[f(y) < f(x)]$ is bounded below by a constant (Lemma 3.3), so the ratio bound is all that is needed.

3. CASE D NEAR C_{BFT}

Lemma 3.1 ([KSFT, Cor. 2.2]). *If $\delta(P) \leq 0.2764$ and $h(x) \leq h(y)$, then $\mathbb{P}[f(y) < f(x)] \leq 0.2764$.*

Proof. If $x \prec y$ the probability is 0. Otherwise suppose $\mathbb{P}[f(y) < f(x)] > 0.2764$. By Fact I1, $\mathbb{P}[f(x) < f(y)] \geq e^{-1} > 0.2764$. Then $\delta(P; x, y) > 0.2764$, which is a contradiction. $\square$

Lemma 3.2 ([KSFT, Lemma 2.4]). *In Case D, with the notation of Fact I4 and $R := S - 2a_1 - a_2$ (≥ 0):*

- (a) $S/2 \geq C_{\text{BFT}} + R/(5 + 3\sqrt{5})$. *This part needs no hypothesis on $\delta(P)$.*
- (b) $R \geq \mathbb{P}[f(x) - f(y) \geq 2] + \mathbb{P}[f(y) - f(z) \geq 2]$.
- (c) *If $\delta(P) \leq 0.2764$, then $\delta(P; x, y) = p$, $\delta(P; y, z) = p'$, and $\max(p, p') \geq C_{\text{BFT}} + R/(5 + 3\sqrt{5})$.*

Proof. (a) Since $S > 0$, put $b_i = a_i/S$ and $s = 1 - R/S = 2b_1 + b_2 \in [0, 1]$. Fact I4 gives $b_2 + b_3 \leq \frac{1}{2}$ and $4b_1b_3 \leq b_2^2$. Let $F(s) = \max(\frac{7}{2}s, \frac{7}{2}\sqrt{s^2 + 1/4} - \frac{1}{4})$. We claim $7b_1 + 3b_2 + 3b_3 \leq F(s)$. If $b_1 = 0$, the left side is $3(b_2 + b_3) \leq 3/2 \leq F(s)$. If $u := b_1 > 0$, then $u \leq s/2$ and $b_2 + b_3 \leq b_2 + b_2^2/(4u) = s^2/(4u) - u$, so the left side is at most $7u + 3 \min(\frac{1}{2}, s^2/(4u) - u)$. The two terms in the minimum are equal at $u_0 = (\sqrt{1 + 4s^2} - 1)/4 \leq s/2$. On $(0, u_0]$ the bound is $7u + \frac{3}{2}$, which is increasing, with value $\frac{7}{2}\sqrt{s^2 + 1/4} - \frac{1}{4}$ at u_0 . On $[u_0, s/2]$ it is $4u + 3s^2/(4u)$, which is convex, so its maximum is at an endpoint; the value at $s/2$ is $\frac{7}{2}s$. Since F is convex with $F(0) = \frac{3}{2}$ and $F(1) = c := (7\sqrt{5} - 1)/4$, we get $F(s) \leq c - (c - \frac{3}{2})(1 - s)$. Hence

$$7 \leq 9S + S(7b_1 + 3b_2 + 3b_3) \leq S(9 + F(s)) \leq (9 + c)S - (c - \frac{3}{2})R.$$

The claim follows from $7/(2(9 + c)) = C_{\text{BFT}}$ and $(c - \frac{3}{2})/(2(9 + c)) = 1/(5 + 3\sqrt{5})$.

(b) Swapping adjacent incomparable elements gives $p(-1, j) = p(1, j - 1)$ and $p(j, -1) = p(j - 1, 1)$ for $j \geq 2$. Hence $2a_1 + a_2 = p(-1, 2) + p(-1, 3) + p(2, -1) + p(3, -1)$. Since $x \prec z$, the event $\{f(x) = f(y) + 1\}$ forces $f(z) - f(y) \geq 2$, so $\{f(x) = f(y) + 1\} = \bigsqcup_{j \geq 2} \{f(y) - f(x) = -1, f(z) - f(y) = j\}$. Therefore

$p - \mathbb{P}[f(x) - f(y) \geq 2] \geq p(-1, 2) + p(-1, 3)$. Symmetrically, $p' - \mathbb{P}[f(y) - f(z) \geq 2] \geq p(2, -1) + p(3, -1)$. Adding the two gives (b).

(c) By Lemma 3.1, $p, p' \leq 0.2764 < \frac{1}{2}$. So $\delta(P; x, y) = p$, $\delta(P; y, z) = p'$, and $\max(p, p') \geq S/2$. Now apply (a). $\square$

Lemma 3.3 (the window forces balance). *In Case D, if $\delta(P) \leq C_{\text{BFT}} + \theta$ with $0 \leq \theta \leq \theta_0$, then $p \geq C_{\text{BFT}} - \theta$ and $p' \geq C_{\text{BFT}} - \theta$.*

Proof. By Lemma 3.2(a), $S \geq 2C_{\text{BFT}}$. By Lemma 3.2(c), $p' \leq \delta(P) \leq C_{\text{BFT}} + \theta$. Hence $p \geq 2C_{\text{BFT}} - p' \geq C_{\text{BFT}} - \theta$, and symmetrically for p' . $\square$

4. STRUCTURE OF BFT TRIPLES IN THE WINDOW

Lemma 4.1 ([KSBFT, Lemma 3.2] under a weaker hypothesis). *Let $\pi(P) \leq D$ and $\delta(P) \leq C_{\text{BFT}} + \theta$, where $0 \leq \theta \leq \theta_0$ and $\theta < \theta_W(D)$. Let (x, y, z) be any BFT triple. Then: (i) it is in Case D; (ii) z covers x ; (iii) y is incomparable only to x and z ; (iv) with $L = \{w : w \prec y\}$ and $U = \{w : y \prec w\}$, we have $X = L \sqcup \{x, y, z\} \sqcup U$, and $L \prec z$, $x \prec U$, $L \prec U$.*

Proof. (i) We have $\delta(P) \leq C_{\text{BFT}} + \theta_0 = 0.2764$. Cases A and B, and their duals, give a pair with $\delta \geq 1/3$, by Facts I2 and I1 (note $1 - e^{-1} < 2/3$). Case C gives a pair with $\delta > 0.2764$ by Fact I3. Both contradict $\delta(P) \leq 0.2764$.

(ii) Suppose $x \prec w \prec z$. Then $g(z) - g(x) \geq 2$ for every g . The event $f(x) < f(y) < f(z)$ has positive probability, since adding $x < y < z$ creates no cycle. On that event $g(z) - g(x) \geq 3$. So $h(z) - h(x) > 2$, which is a contradiction.

(iii) By Lemma 3.2(c) and the hypothesis, $R \leq (5 + 3\sqrt{5})\theta$. Suppose some g has $g(y) < g(w) < g(x)$. Then $N(x, y) \neq \emptyset$ by Theorem 1.2(a). Theorem 1.2(b), Lemma 3.3 and Lemma 3.2(b) then give

$$\frac{C_{\text{BFT}} - \theta}{D + 1} \leq \frac{p}{\pi_N + 1} \leq \mathbb{P}[f(x) - f(y) \geq 2] \leq R \leq (5 + 3\sqrt{5})\theta.$$

This is equivalent to $\theta \geq \theta_W(D)$, a contradiction. The same argument applies to the ordered pair (y, z) : use Theorem 1.2 with (x, y) replaced by (y, z) , whose denominator is $p' = \mathbb{P}[f(z) < f(y)] \geq C_{\text{BFT}} - \theta$, together with the second summand of Lemma 3.2(b). Hence every g satisfies

$$(1) \quad g(y) < g(x) \Rightarrow g(x) = g(y) + 1, \quad g(z) < g(y) \Rightarrow g(y) = g(z) + 1.$$

By Theorem 1.2(a), (1) says $N(x, y) = N(y, z) = \emptyset$. In other words, every $w \neq x, y$ has $w \prec y$ or $x \prec w$, and every $w \neq y, z$ has $w \prec z$ or $y \prec w$. If some $w \notin \{x, y, z\}$ were incomparable to y , this would give $x \prec w \prec z$, contradicting (ii).

(iv) The partition follows from (iii). For $w \in L$, since $w \not\prec y$, we get $w \prec z$. For $w \in U$, $x \prec w$. Finally, $L \prec U$ holds through y . $\square$

Remark 4.2. In [KSBFT, Lemma 3.2(iii)] the chain is $q_3 \leq \mathbb{P}[E] \leq \mathbb{P}[f(x) - f(y) \geq 2] \leq R$, with $E = \{f(y) < f(w) < f(x)\}$ and $q_3 = (D + 1)^{-3(D+1)}$ from their Lemma 3.1. The proof above bounds the *same* middle term, for the same ordered pair, in the same orientation and for the same event as in Lemma 3.2(b), and it does so directly. So q_3 and $\mathbb{P}[E]$ disappear. This is the only point where the exponential loss entered.

5. THE DISPLACEMENT ARGUMENT WITH LINEAR LOSSES

Let $v_1, \dots, v_n$ list X in non-decreasing order of h , let $d_i = h(v_i) - i$, and let $M = \max_i |d_i|$.

Lemma 5.1 ([KSBFT, Lemma 4.1]). *Suppose $G(P)$, the incomparability graph, is connected and $n \geq 3$. Then some k has (v_k, v_{k+1}, v_{k+2}) a BFT triple with $\max(|d_k|, |d_{k+2}|) = M$.*

Proof. Positions are distinct integers, so $d_1 \geq 0$, $d_1 + d_2 \geq 0$, $d_n \leq 0$ and $d_{n-1} + d_n \leq 0$. Hence either $d_i = M$ for some $i \leq n - 2$, or $d_j = -M$ for some $j \geq 3$. Take $k = i$ in the first case and $k = j - 2$ in the second. Then $h(v_{k+2}) - h(v_k) = 2 + d_{k+2} - d_k \leq 2$. Suppose the triple were a chain $x \prec y \prec z$. Then $g(z) - g(x) = 2$ for every g . Any w incomparable to one of x, y, z satisfies $w \not\prec x$ and $z \not\prec w$, so some g has $g(x) < g(w) < g(z)$, which gives $g(z) - g(x) \geq 3$. That is a contradiction. So the triple would be an isolated set in $G(P)$, which contradicts connectedness. $\square$

Proposition 5.2 (mg-d707, Prop. B). *If $G(P)$ is connected, $n \geq 2$ and $\pi(P) \leq D$, then $M \geq d_1 \geq 1/(D+1)$. This is sharp for a D -chain plus one isolated point.*

Proof. The element $a = v_1$ is minimal, since $w \prec a$ would give $h(w) < h(a)$. Because $G(P)$ is connected, a has an incomparable element. For g with $g(a) = 1$, let c be the first element incomparable to a , at position $t \geq 2$. Each element u in positions $2, \dots, t-1$ satisfies $a \prec u$. Then $u \not\prec c$, since otherwise $a \prec c$; and $c \not\prec u$ by position. So $u \parallel c$. Moving c to the front therefore gives an extension with c, a in positions 1, 2. The original g is recovered from the image and from t . Since c passed $t-1$ elements incomparable to it, $t-1 \leq \pi(c) \leq D$. Hence $\#\{g(a) = 1\} \leq D \#\{g(a) = 2\}$. This gives $\mathbb{P}[f(a) = 1] \leq D/(D+1)$ and $d_1 = \mathbb{E}f(a) - 1 \geq \mathbb{P}[f(a) \geq 2] \geq 1/(D+1)$. $\square$

Proposition 5.3 (mg-d707, Prop. C; replaces [KSFT, Lemma 4.2]). *Assume the conclusions of Lemma 4.1 for the triple $(x, y, z) = (v_k, v_{k+1}, v_{k+2})$ of Lemma 5.1. Then*

$$a := \max(\delta(P; x, y), \delta(P; y, z)) \geq C_{\text{BFT}} + \frac{M}{20 + 2\sqrt{5}}.$$

Proof. Height identities (after [KSFT, (4.2)–(4.7)]). Let $L' = L \cup \{x\}$ and $U' = U \cup \{z\}$. Then $L' \prec U'$. Every $w \in L$ has $h(w) < h(y)$ and $w \neq x$, so it precedes x in the h -order; symmetrically for U . Hence $x = v_m$ with $m = |L'|$ (so $k = m$). In every linear extension, L' occupies the first m positions (except that y may interleave) and U' the last. The element y sits between the two blocks, or immediately before x when x is last in the L' -block, or immediately after z when z is first in the U' -block.

Let $f_{L'}$ and $f_{U'}$ be independent uniform extensions of $P[L']$ and $P[U']$. Put $r = \mathbb{P}[f_{L'}(x) = m]$, $s = \mathbb{P}[f_{U'}(z) = 1]$, $v = r + s$ and $Z = 1 + v$. Then $e(P) = e(L')e(U')Z$, so $p = r/Z$ and $p' = s/Z$, both at most $\frac{1}{2}$. Hence $a = \max(r, s)/Z$.

Let α be the expected number of elements of L after x , and β the expected number of elements of U before z . Counting the positions of y gives $\alpha = (1+s)t/Z$ with $t = \mathbb{E}[m - f_{L'}(x)] \geq 1 - r$, and similarly $\beta = (1+r)u/Z$ with $u \geq 1 - s$. Counting the elements below x and z gives $h(x) = m + p - \alpha$ and $h(z) = m + 2 + \beta - p'$. Hence $d_m = p - \alpha$ and $d_{m+2} = \beta - p'$. The condition $h(z) - h(x) \leq 2$ gives $\alpha + \beta \leq v/Z$. Put $\alpha_* = \alpha - (1+s)(1-r)/Z \geq 0$ and $\beta_* = \beta - (1+r)(1-s)/Z \geq 0$. Then

$$0 \leq \alpha_* + \beta_* \leq g/Z, \quad g := v + 2rs - 2 \geq 0.$$

Linear conversion. Let $\rho = (\sqrt{5} - 1)/2$, so $\rho^2 = 1 - \rho$, $1 + 2\rho = \sqrt{5}$ and $C_{\text{BFT}} = \rho/\sqrt{5}$. Write $r = \rho + X$ and $s = \rho + Y$. Since $r, s \in [0, 1]$, we have $|X|, |Y| \leq \rho$. Put $W = |X| + |Y|$ and $\varepsilon = a - C_{\text{BFT}}$.

- (1) $r/Z - C_{\text{BFT}} = A/(\sqrt{5}Z)$ with $A = (1+\rho)X - \rho Y$. Symmetrically $s/Z - C_{\text{BFT}} = B/(\sqrt{5}Z)$ with $B = (1+\rho)Y - \rho X$. So $\sqrt{5}Z\varepsilon = \max(A, B)$, with $A + B = X + Y$ and $A - B = \sqrt{5}(X - Y)$.
- (2) $g = \sqrt{5}(X + Y) + 2XY$, using $2\rho + 2\rho^2 = 2$. We show $X + Y \geq 0$. Suppose $X + Y < 0$. If $XY \leq 0$, then $g < 0$. If $XY > 0$, both are negative and $2|X||Y| \leq 2\rho|Y| < \sqrt{5}W$, so again $g < 0$. Either way this contradicts $g \geq 0$.
- (3) $\max(A, B) = \frac{X+Y}{2} + \frac{\sqrt{5}|X-Y|}{2} \geq \max(X, Y) \geq W/2$, where the last step uses $X + Y \geq 0$. Hence $W \leq 2\sqrt{5}Z\varepsilon$, and in particular $\varepsilon \geq 0$.
- (4) We have $d_m = F/Z - \alpha_*$ and $d_{m+2} = G/Z + \beta_*$, where
 - $F = r - (1+s)(1-r) = (2+\rho)X - (1-\rho)Y + XY$;
 - $G = (1+r)(1-s) - s = (1-\rho)X - (2+\rho)Y - XY$.

Using $|XY| \leq \rho \min(|X|, |Y|)$, both satisfy $|F|, |G| \leq (2+\rho)W$. Also $2|X||Y| \leq \rho W$ gives $g \leq (\sqrt{5} + \rho)W$. Hence $|d_m|, |d_{m+2}| \leq (2+2\rho + \sqrt{5})W/Z = (1+2\sqrt{5})W/Z \leq (20+2\sqrt{5})\varepsilon$.

By the choice of the triple, $M = \max(|d_m|, |d_{m+2}|) \leq (20+2\sqrt{5})\varepsilon$. $\square$

6. PROOF OF THEOREM 1.3

The components of $G(P)$ are totally ordered by $\prec$. Indeed, if $a \prec b$ lie in different components and $a' \parallel a$, then a' is comparable to b , and $b \prec a'$ would give $a \prec a'$; so $a' \prec b$, and connectedness propagates this. So each component K is an interval of every linear extension, and probabilities of events inside K are the same in P and in $P[K]$. Hence $\delta(P) \geq \delta(P[K])$, and $\pi(P[K]) \leq D$. Since P is not a chain, some K has $|K| \geq 2$. If $|K| = 2$, then $\delta(P) \geq \frac{1}{2}$. So we may assume that $G(P)$ is connected and $n \geq 3$.

Suppose $\delta(P) \leq C_{\text{BFT}} + \theta$, with $0 \leq \theta \leq \theta_0$ and $\theta < \theta_W(D)$. Lemma 5.1 gives a BFT triple with $\max(|d_k|, |d_{k+2}|) = M$. Lemma 4.1 applies to it, and Propositions 5.3 and 5.2 give

$$\delta(P) \geq a \geq C_{\text{BFT}} + \frac{M}{20 + 2\sqrt{5}} \geq C_{\text{BFT}} + \frac{1}{(20 + 2\sqrt{5})(D + 1)}.$$

But $\theta < \theta_W(D) < \frac{C_{\text{BFT}}}{(5+3\sqrt{5})(D+1)} = \frac{0.02360\dots}{D+1} < \frac{0.04086\dots}{D+1} = \frac{1}{(20+2\sqrt{5})(D+1)}$, which contradicts $\delta(P) \leq C_{\text{BFT}} + \theta$. If $\theta_W(D) > \theta_0$, take $\theta = \theta_0$. Otherwise let $\theta \uparrow \theta_W(D)$. $\square$

Numbers (from `code/ksbft_k_3c5b/appA_check.py`):

- $\theta_0 = 6.797750 \cdot 10^{-6}$;
- $\theta_W(3471) = 6.799026 \cdot 10^{-6} > \theta_0 > \theta_W(3472) = 6.797068 \cdot 10^{-6}$, so $D_0 = 3471$;
- at $D = 10^4$ the gain is $2.36 \cdot 10^{-6}$, and at $D = 10^6$ it is $2.36 \cdot 10^{-8}$.

Remark 6.1 (which constraint binds). Three constraints appear in the proof:

- the cap θ_0 , which comes only from Case C (Fact I3) via Lemma 4.1(i) and Lemma 3.1;
- Lemma W's $\theta_W(D)$;
- the witness term $1/((20 + 2\sqrt{5})(D + 1))$.

The witness term never binds. The cap θ_0 is independent of D and no strengthening of Lemma W removes it. By Example 2.1, the constant $1/(D + 1)$ cannot be improved using only "Case-D BFT triple + range $\leq D$ ". Whether it improves under the global hypothesis $\delta(P) \leq C_{\text{BFT}} + \theta$ is open. That hypothesis is vacuous for $D \leq 3471$ by the theorem itself. If Lemma 4.2's $M^2/441$ were used in place of Proposition 5.3, the D -uniform range would end at $D = 17$.

Remark 6.2 (where finiteness enters). The bound cannot hold for infinite posets. [BFT95, Thm. 1.4] gives thin infinite posets of range 2 (for example the Fibonacci poset, $x_i < x_j$ iff $j - i \geq 2$) whose balance constant is exactly C_{BFT} . Lemma W and Lemma 4.1 pass to such limits. At the central triples of F_m we have $N(x, y) = \emptyset$ and the bridged $2 + 1$ structure holds. Finiteness is used only in the witness step. Lemma 5.1 needs the boundary relations $d_1 \geq 0$, $d_1 + d_2 \geq 0$, $d_{n-1} + d_n \leq 0$, and Proposition 5.2 needs a bottom element v_1 with $d_1 \geq 1/(D + 1)$. In F_m the displacement M is attained at the ends and stays bounded away from 0, while every fixed central displacement tends to 0.

Corollary 6.3 (effect on [KSBFT, Thm. 1.2]). *In [KSBFT], Theorem 1.2 ($\delta(P) \geq C_{\text{BFT}} + \epsilon$ for all finite non-chains) is obtained by combining Thm. 1.3 at $D = L$ with Thms. 1.4 and 1.5, where L is the range threshold of Thm. 1.5 from [AK25b]. Using Theorem 1.3 in place of Thm. 1.3 gives $\epsilon = \min(\theta_0, \theta_W(L))$, which is about $0.0236/(L + 1)$ once $L > 3471$. The double exponential in L disappears, and the size of ϵ is governed by L alone. The paper does not print L [KSBFT, p. 5], and it omits the derivation of its (1.5), so we give no numerical value. The conditionality of Thm. 1.4 on [AK25a] and Haqi's preprint is unchanged.*

7. ON FACT I3: APPENDIX A OF [KSBFT] AND THE VALUE 67/242

We checked the proof of [KSBFT, Lemma 2.3] (Appendix A) against the preprint of [BFT95]. It imports four results:

- Fact I1;
- [BFT95, Lemma 2.2], the packing lemma, which gives $h(y) - h(x) \geq H(p, t)$ for the packed height function H ;
- [BFT95, Lemma 2.3], the monotonicity lemma: $H(B, \epsilon)$ is decreasing in ϵ for $B \leq 1/3$, except in case (ii) with $k = 1$ and $\epsilon \leq 1/\sqrt{2}$, which occurs only when $B \geq 1 - 1/\sqrt{2} \approx 0.293$;
- [BFT95, Lemma 6.2] (via Lemma 6.1): if x, y, z are pairwise incomparable and $h(z) \leq h(x) + 2$, then $\mathbb{P}[f(x) - f(y) \geq 2] + \mathbb{P}[f(y) - f(z) \geq 2] \geq 1/11$. This needs no bound on p or p' .

Everything else in the appendix is algebra, which we re-derived by hand and checked numerically:

- the closed forms (A.2) for H , which agree with the definition of packed sequences to $3 \cdot 10^{-13}$ on a grid;
- Claim A.1;
- the boundary identity and bound (A.5);
- both sub-cases of Claim A.2, including the packed-case conditions $1 + 2t + 2t^2 \leq 1/p \leq (1 + t)(1 + 2t)$;
- the final inequality, $5 - 11 \cdot 0.2764 + \frac{1}{22} = 2.00505 > 2$.

One stated justification is too weak but the conclusion holds. Appendix A derives $3/5 < v_q < 1$ from $q \in (1/5, 3/10)$, which is false near $q = 3/10$ ($v_{3/10} \approx 0.565$). The bound does hold on the range actually established, $q \in [6/11 - 0.2764, 0.2764]$, where $v_q \in [0.618, 0.636]$.

The value 67/242. The last line of the appendix reads $5 - \frac{11}{2}(p + p') + \frac{1}{22} > 2$, and it holds for every threshold $T < 67/242$. Every other step (Claim A.1's use of $T < 1 - 1/\sqrt{2}$, the range of v_q , Claim A.2's $32/9 < 1/T$, and the first branch, which needs $T < 37/132$) remains valid at $T = 67/242$. So the argument proves $\max(\delta(P; x, y), \delta(P; y, z)) \geq 67/242 \approx 0.276860$, with a non-strict inequality: at $T = 67/242$ the chain gives exactly $h(z) - h(x) \geq 2$. The appendix's claim that it “yields the stronger lower bound 67/242” is therefore correct, modulo the same imports. With it, Lemma 4.1(i) holds for every $\theta < \theta'_0 := 67/242 - C_{\text{BFT}} = 4.663 \cdot 10^{-4}$. Lemma 3.1 and Lemma 3.2(c) hold at threshold $67/242 < e^{-1}$ verbatim. Theorem 1.3 then improves to

$$\delta(P) \geq C_{\text{BFT}} + \min(\theta'_0, \theta_W(D)),$$

which is $\geq 67/242$ for $D \leq 49$.

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